Cambridge
O Level

**Cambridge Assessment International Education**
Cambridge Ordinary Level

**BIOLOGY** **5090/32**

Paper 3 Practical Test **October/November 2018**

MARK SCHEME

Maximum Mark: 40

**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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## Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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| GENERIC MARKING PRINCIPLE 5:<br><br>Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen). |
| GENERIC MARKING PRINCIPLE 6:<br><br>Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind. |

Mark schemes will use these abbreviations:

| | |
|---|---|
| **;** | separates marking points |
| **/** | alternatives |
| **()** | contents of brackets are not required but should be implied |
| **R** | reject |
| **A** | accept (for answers correctly cued by the question, or guidance for examiners) |
| **Ig** | ignore (for incorrect but irrelevant responses) |
| **AW** | alternative wording (where responses vary more than usual) |
| **AVP** | alternative valid point (where a greater than usual variety of responses is expected) |
| **ORA** | or reverse argument |
| **underline** | actual word underlined must be used by candidate |
| **+** | statements on both sides of the **+** are needed for that mark |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 1(a)(i) | milk **+** yoghurt-milk mixture liquid / fluid / runny / watery **;**<br><br>yoghurt thick(er) / not very liquid / not fluid / not runny **;**<br><br>all samples white / same colour **;** | **3** | |
| 1(a)(ii) | *milk pH*: 6 / 7 / 8 **;**<br>*yoghurt-milk mixture pH*: 6 / 7 / 8 **;**<br>*yoghurt pH*: 3 / 4 / 5 **;** | **3** | |
| 1(a)(iii) | decreases / falls **AW ;** | **1** | **A** becomes acidic / increases in acidity |
| 1(b)(i) | kill **or** destroy bacteria / microorganisms present **;** | **1** | |
| 1(b)(ii) | thermometer **;** | **1** | **A** temperature sensor |
| 1(b)(iii) | pH has reached stated value in candidate's results **;**<br><br>consistency **or** texture is thick / creamy **;** | **2** | |

| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 2(a)(i) | table completed **+** ticks and crosses used **;**<br>index finger and thumb same number of ticks (±1) **;**<br>palm less sensitive than finger **;**<br>wrist less sensitive than finger **;** | **4** | |
| 2(a)(ii) | answer consistent with candidate's results **;** | **1** | |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 2(b)(i) | 1 both axes fully labelled with units on distance axis ;<br><br>2 linear scale with 0 at origin for mean minimum distance **+** at least half of grid used ;<br><br>3 all data correctly plotted ±1 mm ;<br><br>4 all drawn bars ruled and of equal width ; | **4** | |
| 2(b)(ii) | to make results more <u>reliable</u> / improve <u>reliability</u> ; | **1** | |
| 2(b)(iii) | prevent seeing / knowing how many points being tested ;<br><br>so that touch is the only sense being tested ; | **1** | |
| 2(b)(iv) | ensure same area of each section of arm / shoulder tested ;<br><br>same pressure / force exerted on skin / time points in contact with skin ;<br><br>toothpicks of equal sharpness / same surface area ; | **1** | |
| 2(b)(v) | different areas of skin have different numbers of receptors ;<br><br>(touch) receptors closer together in more sensitive areas / hands **ORA** ;<br><br>fingers / hand involved in manipulating / handling objects **ORA** ; | **2** | **A** more (touch) receptors in finger tip |

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| Question | Answer | Marks | Guidance |
|---|---|---|---|
| 3(a)(i) | 1 drawing of pod at least 80 mm in length ;<br><br>2 outlines of three peas drawn with sharp pencil **+** continuous lines with stalks delimited (if stalks have been drawn) **+** no shading anywhere ;<br><br>3 <u>all</u> peas drawn roughly circular **+** not touching lower edge of pod ;<br><br>4 all 3 stalks joining peas to pod drawn with double lines ;<br><br>5 seed and pericarp / fruit wall correctly labelled ; | **5** | |
| 3(a)(ii) | measurement of photo: 59 – 61 ;<br><br>candidate's measurement from photograph ÷ 1.2 ;<br><br>correct answer with units ; | **3** | correct answer with units but no working scores both marks 2 and 3 |
| 3(b)(i) | <u>24 round + 6 wrinkled</u> ; | **1** | |
| 3(b)(ii) | 6 / 30 (×100) ;<br><br>20 ; | **2** | correct answer with no working scores 2 marks |
| 3(b)(iii) | actual ratio = 4:1 ;<br><br>sample size too small **AW** ;<br><br>random fertilisation / based on chance **or** probability ; | **2** | |
| 3(c) | Biuret solution ;<br><br>blue **+** change to mauve / purple / lilac indicates protein ; | **2** | **R** if heated |

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